Project 25-26: Learning to Defer with Neuro-Symbolic Hierarchical Concepts

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Abstract

Learning to Defer (L2D) allows machine learning models to dynamically decide whether to predict a task themselves or defer to a human expert [1]. Deferring Concept Bottleneck Models (DCBMs) [2] extend this framework by allowing deferral at both the concept and task levels, under the assumption that concepts are independent. However, in many real-world scenarios, concepts are correlated one to another (e.g., if the traffic light is green it cannot be red). This project aims to extend DCBMs with a Neuro-Symbolic Hierarchical Concept Layer, enabling the model to reason over concept dependencies. We hypothesize that incorporating hierarchical structure can reduce unnecessary deferrals, improve task accuracy, and provide more informative explanations for human-AI collaboration.

1 Defer in Concept Bottleneck Model

Let $\mathcal{X}$ denote the input space, $\mathcal{C}$ the concept space, and $\mathcal{Y}$ the task labels. A standard Concept Bottleneck Model (CBM) decomposes prediction into two stages:

$$f : \mathcal{X} \to l_{\mathcal{C}}, \qquad \omega : l_{\mathcal{C}} \to \Delta_{\mathcal{Y}}, \tag{1}$$

where f is the concept extractor predicting logits for concepts $\mathbf{C}$ from input $\mathbf{X}$, and ω is the task predictor that maps concepts logits to task labels $\mathbf{Y}$. A deferring system is then added for each concept and task variable:

$$\Gamma_{\mathbf{C}} = \begin{cases} M_{\mathbf{C}}(\mathbf{x}), & \text{predict directly,} \\ H_{\mathbf{C}}, & \text{defer to human,} \end{cases} \tag{2}$$

DCBMs assume concept independence, i.e., $\mathbf{C}_i \perp \mathbf{C}_j$ for $i \neq j$, which simplifies the surrogate loss used to train the deferring system [2]. When this assumption is violated, e.g., concepts are organized in a hierarchy, independent deferring decisions may be suboptimal: correcting a parent concept might render child concepts unnecessary to defer, or introduce inconsistencies if deferrals are made independently.

2 Hierarchical Neuro-Symbolic Classification

Hierarchical classification addresses learning problems in which labels are organized according to a hierarchy (typically a tree or a directed acyclic graph), and each instance may be associated with multiple labels that must respect hierarchical constraints. Formally, let

$$\mathcal{H} = (\mathcal{Y}, \preceq) \tag{3}$$

be a hierarchy over a finite label set $\mathcal{Y}$, where $\preceq$ denotes the ancestor–descendant relation. Given an input space $\mathcal{X}$, the goal of Hierarchical Multi-Label Classification (HMC) is to learn a function

$$f : \mathcal{X} \to 2^{\mathcal{Y}} \tag{4}$$

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such that for every predicted label $y \in f(x)$, all ancestors of $y \in (\mathcal{H}$ are also included in $f(x)$.

Neuro-symbolic approaches integrate sub-symbolic learning (neural networks) with symbolic reasoning to enforce such structural constraints.

An example of such approach is **Coherent Hierarchical Multi-Label Classification Networks (CHMC)** [3]. In CHMC, a neural classifier produces initial confidence scores for each label, which are then projected onto the feasible set of hierarchically consistent label assignments using *fuzzy logic*. This layer is completely differentiable, so you can rely on cross-entropy for classification purposes. This guarantees coherence of predictions with respect to the hierarchy while preserving the expressive power of deep learning.

More recently, **Semantic Probabilistic Layer** [4] have been introduced. The idea is similar to CHMC, namely to directly embed logical constraints into neural architectures. SPL provides a differentiable *probabilistic reasoning layer* that enforces logical and hierarchical relations during both training and inference.

3 Plan

- Study the foundational papers:
  - Learning to Defer [1];
  - Deferring CBMs [2]; Code for DCBM,
  - Neuro-Symbolic Hierarchical Classification [3, 4]. Code for CHMC, code for SPL.
- Select a dataset with dependencies among concepts (e.g., CIFAR-100 with a constructed hierarchy, or a shark-related dataset with semantic structure).
- Implement the Hierarchical-Concepts DCBM (H-DCBM) by designing hierarchical constraints over the concepts and integrating a neuro-symbolic layer on top of the concept layer.
- Replicate the standard DCBM settings and, leveraging the hierarchical structure, allow deferral only at the level of leaf concepts.
- Compare H-DCBMs with standard DCBMs on benchmark datasets.
- Evaluate whether hierarchical reasoning reduces the number of deferrals, improves task accuracy, and provides interpretable explanations for deferred decisions.

References

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